

Answers to exercise 04

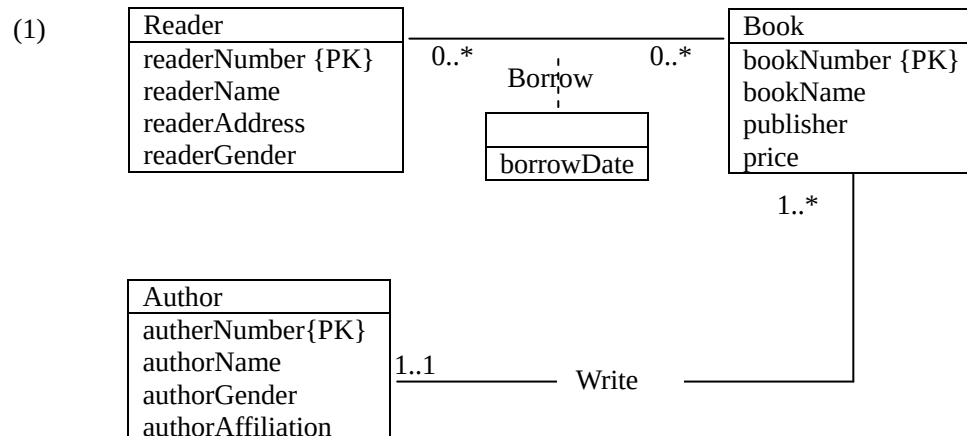
PART A

1	C	2	B	3	A	4	B	5	C
6	A	7	B	8	C	9	C	10	B
11	A	12	C	13	C	14	B	15	A
16	B								

PART B

- (a) database planning, (b) system definition, (c) requirements collection and analysis, (d) database conceptual design, (e) database logical design, (f) database physical design, (g) implementation, (h) data conversion and loading, (i) testing, (j) operational maintenance
- (a) conceptual, (b) logical, (c) physical
- (a) objects, (b) properties, (c) rectangle, (d) associations, (e) entity types, (f) relationship type
- (a) primary key, (b) foreign key, (c) a relation, (d) primary keys
- (a) distinct object, (b) property, (c) association

PART C



(2)

Reader(readerNumber, readerName, readerAddress, readerGender)

Primary key: readNumber

Alternate key: None

Foreign key: none

Author(authorNumber, authorName, authorGender, authorAffiliation)

Primary key: authorNumber

Alternate: none

Foreign key: none

Book(bookNumber, bookName, authorNumber, publisher, price)

Primary key: bookNumber

Alternate key: none

Foreign key: authorNumber

Borrow(readerNumber, bookNumber, borrowDate)

Primary key: (readerNumber, bookNumber, borrowDate)

Candidate key: none

Foreign key: readNumber, bookNumber

(3)

create table Reader(

 readerNumber char(10) primary key,

 readerName char(20) not null,

 readerAddress char(20),

 readerGender char(1) not null check (authorGender in ('M','F'))

)

create table Author (

 authorNumber char(10) primary key,

 authorName char(20) not null,

 authorGender char(1) not null check (authorGender in ('M','F')),

 authorAffiliation char(100)

)

create table Book(

 bookNumber char(10) primary key,

 bookName char(20) not null,

 authorNumber char(10) foreign key references Author(authorNumber),

 publisher char(20),

 price money

)

create table Borrow(

 readerNumber char(10) foreign key references Reader(readerNumber),

 bookNumber char(10) foreign key references Book(bookNumber),

 borrowDate datetime,

 primary key (readNumber, bookNumber, borrowDate)

)